Lesson notes · for you

Measure it right

IoT · Level 1 · Lesson 3

Time
45 min

Materials
A ruler or tape measure (a tailor's tape works); Paper and a pencil for your table; Optional: a thermometer, or a bottle rain gauge (an adult cuts the bottle)

Learning goals

- Read a scale carefully: the unit, eye level and what each small step is worth.
- Record a measurement with its number, unit, time and place.
- Spot a reading that can't be right, and say what to do about it.

Activity: Measure it three times

1. Pick something you can measure several times: your desk with a ruler or tape, the shade temperature with a thermometer, or rain in a bottle gauge.
2. Take two numbered marks, subtract, and divide by the gaps between them. Write the answer at the top of your page.
3. Bring your eye level with the mark each time. Use the same tool, the same place and the same method.
4. Make a table with columns for number, unit, time and place, and fill in one row for each reading.
5. Ask a friend to measure the same thing. If you disagree, look for the reason (the unit, eye level, the place) before changing anything.

If you make a bottle rain gauge, an adult cuts the bottle and tapes the sharp edge. If a glass thermometer breaks, don't touch it: tell an adult.

Discussion

- Why might two careful people get slightly different readings?
- When is it fair to leave a reading out, and what must you write down?
- Where in daily life do you see numbers with no unit?

Check yourself

- There are 5 gaps between 0 and 10 on a scale. What is each gap worth?

Try it this week
Measure your desk three times, the same way, and write each reading with its number, unit, time and place.

Worksheet

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Name _____ Class _____ Date _____

1 **The school garden's soil moisture sensor reads 50 %, then exactly 0 % at midnight, then 50 % again an hour later. Nobody watered. What should you do with the 0?**

- ◯ Keep it, mark it as a glitch, and check the sensor's wires
- ◯ Delete it, so the chart looks clean and nobody is confused
- ◯ Water the garden at once: the sensor says it's bone dry

2 **Read Yusuf's rain gauge.**

The marks 0 and 20 mm have 4 small gaps between them, so each gap is worth _____ mm. The water stands 3 gaps above 0, so _____ mm of rain fell.

Word bank: 5 · 15 · 3 · 12

3 **You measure your desk three times and get 70, 71 and 70 cm. That means you made a mistake.**

◯ True ◯ False

4 **Put the steps of keeping a good temperature log in order.**

- ☐ Mark any glitch, keep it, and note why you left it out
- ☐ Look for readings that can't be right
- ☐ Write the number, unit, time and place
- ☐ Read the thermometer with your eye level with the mark
- ☐ Choose one shady place and one time each day

5 **Which of these records could someone compare with next week's?**

☐ 31 ☐ 31 °C, 2 pm Tuesday, back of the classroom ☐ 31 °C, back of the classroom

☐ Very hot, after lunch ☐ Rain: 12 mm since 8 am Monday, school garden gauge

✎ **Think of a time a number without a unit, or a wrong reading, confused someone: a recipe, a price, a test score. What would have made it clear?**